Mir Mahathir Mohammad - Curriculum Vitae

Contact

Location: Salt Lake City, UT **Phone:** +1 385 210 8554 **Email:** mirmahathir1@gmail.com

Portfolio: https://mirmahathir1.github.io/ LinkedIn GitHub Google Scholar

Education

University of Utah, Salt Lake City, Utah
Ph.D. in Computer Science
Kahlert School of Computing | August 2023 – Present
Advisor: Professor El Kindi Rezig

University of Utah, Salt Lake City, Utah
M.S. in Computer Science
Kahlert School of Computing | August 2023 – December 2025
CGPA: 3.978 / 4.0
Advisor: Professor El Kindi Rezig

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
B.S. in Computer Science
April 2017 – April 2022
Advisor: Professor Muhammad Abdullah Adnan

Research Interests

I focus on data preparation in general and data discovery in particular for data lakes. I built SemDisc, the first end-to-end semantic join discovery system using a query-by-example interface. I also explored how discovered relationships can recommend meaningful ways to organize and sort data. I built NLDisc, which turns a user's natural language description of intent into join discovery over data lakes, and co-developed LORY-J, which treats spatial attributes as first-class signals to discover location-aware joins. I contributed to Buckaroo, a visual data wrangling system that enables users to interactively clean and repair data anomalies in visualizations. My interests include:

Data Systems • Data Lakes • Data Discovery & Integration • Data Wrangling • Data Cleaning • AI for Data Management

Publications

Publications since starting Ph.D. (2023-Present, ~3.1 years)

- ***SIGMOD'26***: Mir Mahathir Mohammad, El Kindi Rezig. "Qualitative Join Discovery in Data Lakes using Examples." Accepted at ACM SIGMOD International Conference on Management of Data (SIGMOD'26), 2026. [PDF] [Link] *a system for discovering hybrid join paths (combining semantic and equi-joins) in data lakes using query-by-example, supporting hidden tables and semantic tuple matching*
- ***VLDB'26 (Demo)***: Lu Xing, Mir Mahathir Mohammad, Walid Aref, El Kindi Rezig. "LORY-J: Location-aware Join Discovery from Data Lakes." Accepted at VLDB 2026 (Demo Track). [PDF] *a system for location-aware join discovery in data lakes that treats spatial attributes as first-class signals, composing hybrid join paths through an interactive, map-driven interface*
- ***SIGMOD'26 (Demo)***: Mir Mahathir Mohammad, El Kindi Rezig. "SemDisc: An End-to-End Query-by-Example Semantic Join Discovery System." Accepted at ACM SIGMOD 2026 (Demo Track). [PDF] [Link] *an interactive demo enabling users to discover hybrid join paths in schema-less data lakes through a query-by-example interface with semantic type annotation and inverted join path indexing*
- ***SIGMOD'26 (Demo)***: Mir Mahathir Mohammad, El Kindi Rezig. "NLDisc: Intent-aware Join Discovery using Natural Language." Accepted at ACM SIGMOD 2026 (Demo Track). [PDF] [Link] *a system for intent-aware join discovery over data lakes using natural language queries with interactive intent disambiguation and join path diversification*
- ***CIDR'26***: El Kindi Rezig, Mir Mahathir Mohammad, Nicolas Baret, Ricardo Mayerhofer, Andrew McNutt, Paul Rosen. "Towards Scalable Visual Data Wrangling via Direct Manipulation." Accepted at CIDR 2026. [PDF] [Link] *a visual data wrangling system that enables users to clean and repair data anomalies through direct manipulation of interactive visualizations*
- ***VLDB'25 (Demo)***: Akash Khatri, Mir Mahathir Mohammad, El Kindi Rezig. "Sort it Like You Mean It: Discovering Semantically Interesting Attribute Augmentations to Sort Tables." Accepted at VLDB 2025 (Demo Track). [PDF] [Link]

Recommends semantically meaningful ways to sort tables by automatically discovering and augmenting attributes from data lakes using LLMs.

Undergraduate research

- ***IEEE FG'24***: Iftekhar E Mahbub Zeeon, Mir Mahathir Mohammad, Muhammad Abdullah Adnan. "BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation." Accepted at IEEE FG 2024. *Introduces the first large-scale sentence-level dataset for Bangla Sign Language translation, derived from 60 hours of YouTube news content with professional signers.* [PDF] [Link]
- ***Neurocomputing'22***: Md. Ashraful Islam, Mir Mahathir Mohammad, Sarkar Snigdha Sarathi Das, Mohammed Eunus Ali. "A survey on deep learning based Point-of-Interest (POI) recommendations." Accepted at Neurocomputing (Journal), 2022. [PDF] [Link] *Categorizes deep learning approaches for POI recommendation systems in location-based social networks.*

Research Experience

University of Utah, Kahlert School of Computing, Salt Lake City, UT
Graduate Research Assistant, August 2023 – Present
Advisor: Professor El Kindi Rezig

- Developed algorithms for qualitative join discovery in data lakes using example-based queries, enabling efficient dataset integration across heterogeneous tabular data
- Built systems for semantic table sorting, natural-language intent-aware join discovery, and location-aware join discovery over data lakes
- Implemented scalable data wrangling systems with direct manipulation interfaces, handling data transformations

Bangladesh University of Engineering and Technology, CSE, Dhaka, Bangladesh
Research Assistant, July 2022 – June 2023
Advisor: Professor Muhammad Abdullah Adnan

- Developed machine learning pipelines for processing and analyzing video datasets for sign language translation
- Built data collection and annotation systems for creating structured datasets, handling data cleaning

Technical Skills

Data Systems & Databases: PostgreSQL, MySQL, MongoDB

Data Processing & ML: PyTorch, TensorFlow

Cloud & Infrastructure: Docker, Google Cloud Platform

Programming Languages: Python, JavaScript/TypeScript, C++, SQL

Development Tools: Node.js, Express.js, React.js, Vue.js, Streamlit

Additional Experience

Everforth, Tokyo, Japan (Remote)
Frontend Developer, April 2022 – June 2023

- Built e-commerce web applications using Vue.js and CakePHP

Past Projects

- **Badhan Blood Donation Management System:** Designed and implemented a full-stack blood donation platform with MongoDB backend, serving users across BUET campus with real-time donor matching and request management [GitHub]
- **Automated Robotic Arm:** Built computer vision and control systems for robotic manipulation tasks using MATLAB, integrating sensor data processing and motion planning algorithms [GitHub]